CHAPTER 2

INFORMATION THEORY

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Learning Objectives for Chapter 2

After studying this module you should understand

• why it is important to measure the amount of uncertainty in a situation of choice;

• why entropy is an appropriate measure of uncertainty;

• how information and uncertainty are related and why therefore entropy plays an important role in measuring information;

• that information is always relative to a precise question and to prior information.

Information Theory

2.1 Introduction

Most scientists agree that information theory began in 1948 with Shannon's famous article. Even if information theory is considered a branch of communication theory, it actually spans a wide number of disciplines including computer science, probability, statistics, economics, etc. The most basic questions treated by information theory are: how can 'information' measured? How can 'information' be transmitted? From a communication theory perspective it is reasonable to assume that the information is carried out either by signals or by symbols. Shannon's sampling theory tells us that if the channel is bandlimited, in place of the signal we can consider its samples without any loss. Therefore, he provided answers to the following questions: What is "information" and how to measure it? And What are the fundamental limits on the storage and the transmission of information? it makes sense to confine the information carriers to discrete sequences of symbols, unless differently stated. This is what we'll do throughout this lectures.

2.2 Model of information transmission system(Shannon Paradigm)

Transmitting a message from a transmitter to a receiver can be sketched as follows:

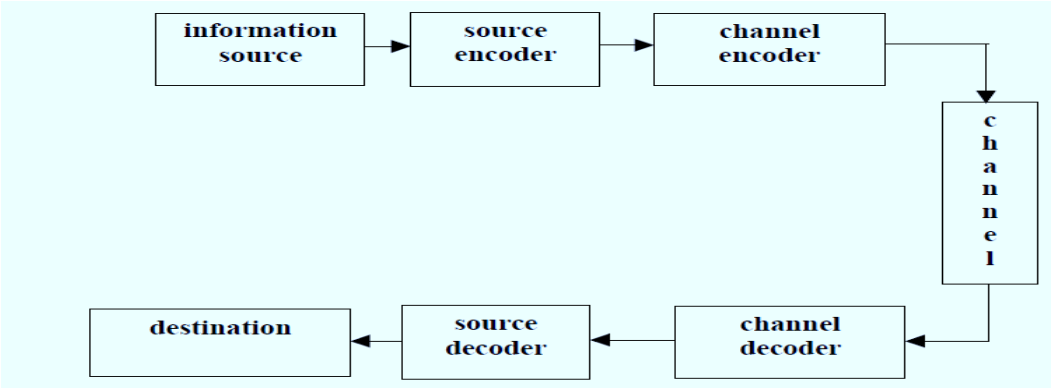


Figure 2.1: Shannon system

The components of information system as described by Shannon are:

1. An information source is a device which randomly delivers symbols from an alphabet. As an example, a PC (Personal Computer) connected to internet is an information source which produces binary digits from the binary alphabet{0, 1}.

2. A channel is a system which links a transmitter to a receiver. It includes signaling equipment and pair of copper wires or coaxial cable or optical fiber, among other possibilities.

3. A source encoder allows one to represent the data source more compactly by eliminating redundancy: it aims to reduce the data rate.

So: Three basic elements to every communication system:

Transmitter: converts message into a form suitable for transmission

Channel: the physical medium, introduces distortion, noise, interference

Receiver: reconstruct a recognizable form of the message

2.3 Self- information

In information theory, **self-information** is a measure of the information content associated with the outcome of a random variable. It is expressed in a unit of information, for example bits, ***nats*****,** or ***hartleys*****,** depending on the base of the logarithm used in its calculation.

A **bit** is the basic unit of information in computing and digital communications. A bit can have only one of two values, and may therefore be physically implemented with a two-state device. These values are most commonly represented as 0 and 1.

A **nat** is the **natural unit of information**, sometimes also **nit** or **nepit**, is a unit of information or entropy, based on natural logarithms and powers of e, rather than the powers of 2 and base 2 logarithms which define the bit. This unit is also known by its unit symbol, the nat.

The **hartley** (symbol **Hart**) is a unit of information defined by International Standard IEC 80000-13 of the International Electrotechnical Commission. One hartley is the information content of an event if the probability of that event occurring is 1/10. It is therefore equal to the information contained in one decimal digit (or dit).

1 Hart ≈ 3.322 Sh ≈ 2.303 nat.

The amount of self-information contained in a probabilistic event depends only on the probability of that event: the smaller its probability, the larger the self-information associated with receiving the information that the event indeed occurred as shown in figure (2.2) which show the relation between the probability and self-information.

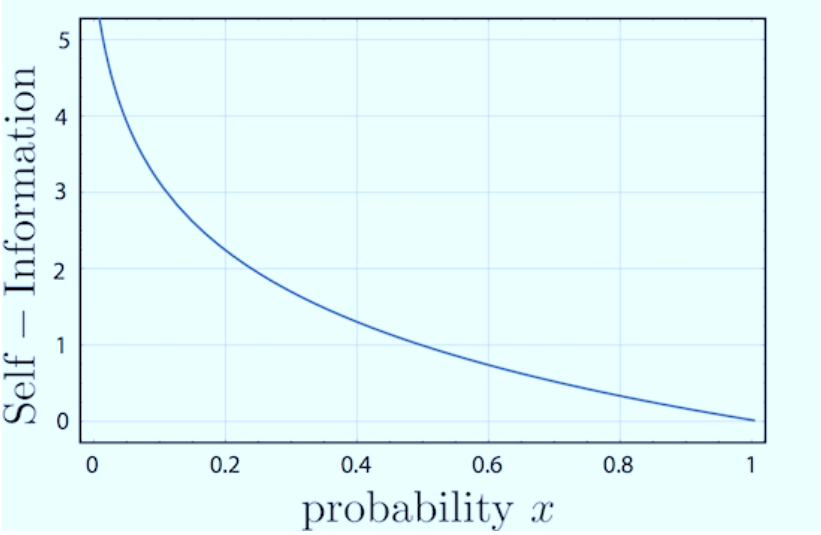


figure (2.2): The relation between the probability and self-information

Suppose that the source of information produces finite set of message $x_1, x_2, \dots \dots . x_n$

with prob. $p(x_1), p(x_2), \dots \dots \dots . P(x_n)$ and such that

$$\sum_{i=1}^{n} P(x_i) = 1$$

1- Information is zero if $P(x_i) = 1$ (certain event)

2- Information increase as $P(x_i)$ decrease to zero

3- Information is a +ve quantity

The log function satisfies all previous three points hence:

$$I(x_i) = -\log_a P(x_i) \quad bit$$

Recall that $\log_a x = \frac{\ln x}{\ln a}$

Where $I(x_i)$ is self information of (x_i) and if:

- If "a" =2 , then $I(x_i)$ has the unit of bits
- If "a"= e = 2.71828, then $I(x_i)$ has the unit of nats
- If "a"= 10, then $I(x_i)$ has the unit of hartly

Example 2.1: A fair die is thrown, find the amount of information gained if you are told that 4 will appear.

Sol:

$$P(1) = P(2) = \cdots \ldots \ldots = P(6) = \frac{1}{6}$$

$$I(4) = -\log_2\left(\frac{1}{6}\right) = \frac{\ln(\frac{1}{6})}{\ln 2} = 2.5849 \text{ bits}$$

Example 2.2: A biased coin has P(Head)=0.3. Find the amount of information gained if you are told that a tail will appear.

Sol:

$$P(\text{tail}) = 1 - P(\text{Head}) = 1 - 0.3 = 0.7$$

$$I(\text{tail}) = -\log_2(0.7) = -\frac{\ln 0.7}{\ln 2} = 0.5145 \text{ bits}$$

H.W: a communication source emits three types of information namely A,B and C ,if these sequence are detected [BBACCBAACC] ,find the number of bits in each symbol of the sequence?

2.4 Average of Information (ENTROPY)

In information theory, **entropy** is the average amount of information contained in each message received. Here, message stands for an event, sample or character drawn from a distribution or data stream. Entropy thus characterizes our uncertainty about our source of information.

2.4.1 Source Entropy H(X)

If the source produces not equi-probable messages then $I(x_i), i = 1, 2, \dots\dots, n$ are different. Then the statistical average of $I(x_i)$ over i will give the average amount of uncertainty associated with source X, this average is called ***source entropy*** and denoted by H(X), given by:

Definition The entropy H(X) of a discrete random variable X is defined by:

$$H(x) = \sum p(x)\, log_2 \frac{1}{p(x)} = -\sum p(x)\, log_2\, p(x) \qquad \frac{bits}{symbol}$$

$$H(X) = \sum_{i=1}^{n} P(x_i)\, I(x_i) \quad \frac{bits}{symbol} \;\gg\gg\; H(X) = -\sum_{i=1}^{n} P(x_i)\, log_a\, P(x_i) \quad \frac{bits}{symbol}$$

For example, the entropy of a fair coin toss is 1 bit. We will use the convention that 0 log 0 = 0, which is easily justified by continuity since x log x → 0 as x → 0.

Example 2.3: Let x={a=1/2, b=1/4 ,c=1/8, d=1/8} , find the average of information?

Sol:

The entropy of x is :

H(x)=-1/2*log 1/2 -1/4* log1/4-1/8*log1/8-1/8*log1/8= 7/8 bits

Example 2.4: Find the entropy of the source producing the following messages:

$$Px_1 = 0.25, \quad Px_2 = 0.1, \quad Px_3 = 0.15, \quad \text{and } Px_4 = 0.5$$

Sol:

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log_a P(x_i)$$

$$= -\frac{[0.25\ln 0.25 + 0.1\ln 0.1 + 0.15\ln 0.15 + 0.5\ln 0.5]}{\ln 2}$$

$$= 1.7427 \ \frac{\text{bits}}{\text{symbol}}$$

2.4.2 Binary Source Entropy $H_b(X)$ or H (X)

In information theory, the **binary entropy function**, denoted or H(X) or Hb(X), is defined as the entropy of a Bernoulli process with probability *p* of one of two values. Mathematically, the Bernoulli trial is modelled as a random variable *X* that can take on only two values: 0 and 1 as:

$P(0_T) + P(1_T) = 1$

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log_a P(x_i)$$

$$H(X) = -\sum_{i=1}^{2} P(x_i) \log_a P(x_i)$$

Example 2.5: Find and plot the entropy of binary source. If $P(0T)$ = 0.2,

Sol:

$$P(0_T) + P(1_T) = 1$$

If $P(0_T) = 0.2$, then $P(1_T) = 1 - 0.2 = 0.8$, and put in equation,

$$H(X) = -[P(0_T)\log_2 P(0_T) + (1 - P(0_T))\ \log_2(1 - P(0_T))] \text{ bits/symbol}$$

$$H(X) = -[0.2\log_2(0.2) + 0.8\log_2(0.8)] = 0.7$$

Not that H(X) is maximum equal to 1(bit) if: $P(0_T) = P(1_T) = 0.5$ as shown in figure(2.3).

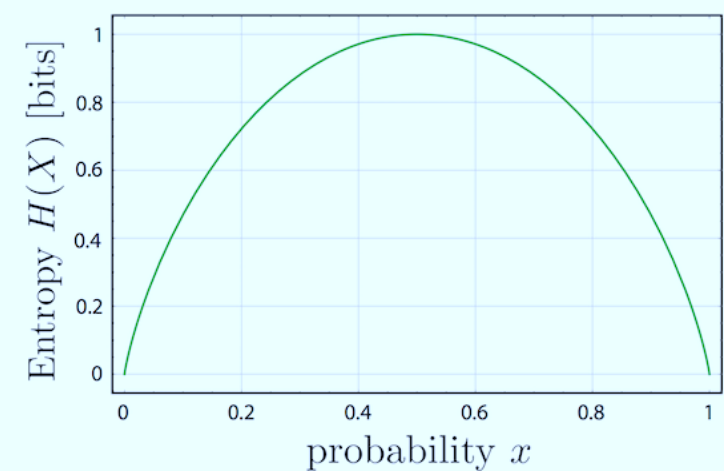


figure (2.3): The relation between the probability and entropy H(x)

2.4.3 Non-Binary Equi-probable Source Entropy H (X)

For any non-binary source, If all messages are equiprobable, then $P(x_i) = 1/n$ so hat:

$$H(X) = H(X)_{max} = -[\frac{1}{n} log_a \left(\frac{1}{n}\right)] \times n = -log_a \left(\frac{1}{n}\right) = log_a n \ \ bits/symbol$$

And $H(X) = 0$ if one of the message has the prob. of a certain event.

Example 2.6: Consider a random variable that has a uniform distribution over 32 outcomes. find the average of information?

$$H(x) = = -\sum_{i=1}^{32} p(x)\log_2 p(x) = \sum_{i=1}^{32} \frac{1}{32}\log_2 \frac{1}{32} = 5\ bits$$

2.4.4 Source Entropy Rate

It is the average rate of amount of information produced per second.

$$R(X) = H(X) \times \text{rate of producing the symbols} \quad = \frac{\text{bits}}{\text{sec}} = \text{bps}$$

The unit of H(X) is bits/symbol and the rate of producing the symbols is symbol/sec, so that the unit of R(X) is bits/sec.

Sometimes $R(X) = \frac{H(X)}{\bar{\tau}}$, $\quad \bar{\tau} = \sum_{i=1}^{n} \tau_i P(x_i)$

Where: $\bar{\tau}$ is the average time duration of symbols

τ_i is the time duration of the symbol x_i.

Example 2.7: A source produces dots '.' And dashes '-' with P(dot)=0.65. If the time duration of dot is 200ms and that for a dash is 800ms. Find the average source entropy rate.

Sol:

$$P(\text{dash}) = 1 - P(\text{dot}) = 1 - 0.65 = 0.35$$

$$H(X) = -[0.65\log_2(0.65) + 0.35\log_2(0.35)] = 0.934 \text{ bits/symbol}$$

$$\bar{\tau} = 0.2 \times 0.65 + 0.8 \times 0.35 = 0.41 \text{ sec}$$

$$R(X) = \frac{H(X)}{\bar{\tau}} = \frac{0.34}{0.41} = 2.278 \text{ bps}$$

H.W: A discrete source emits one of five symbols once every millisecond. The symbol probabilities are 1/2, 1/4, 1/8, 1/16 and 1/16 respectively. Calculate the information rate.

H.W: A source produces dots and dashes; the probability of the dot is twice the probability of the dash. The duration of the dot is 10msec and the duration of the dash is set to three times the duration of the dot. Calculate the source entropy rate.

2.5-Relative Entropy and Mutual Information

We defined the entropy of a single random variable. We now extend the definition to a pair of random variables. There is nothing really new in this definition because (X, Y) can be considered to be a single vector-valued random variable.

2.5.1 Marginal Entropies

Marginal entropies is a term usually used to denote both source entropy H(X) defined as before and the receiver entropy H(Y) given by:

$$H(x) = -\sum_{i=1}^{m} P(xi)\, log_2 P(x_i) \qquad \frac{bits}{symbol}$$

$$H(Y) = -\sum_{j=1}^{m} P(y_j)\, log_2 P(y_j) \qquad \frac{bits}{symbol}$$

2.5.2 Joint entropy H(X, Y) and conditional entropy H(Y | X)and H(X | Y)

The average information associated with the pair (x_i, y_j) is called *joint or system entropy H(X,Y):*

$$H(X,Y) = H(XY) = -\sum_{j=1}^{m}\sum_{i=1}^{n} P(x_i, y_j)\, log_2 P(x_i, y_j) \qquad bits/symbol$$

The average amount of information associated with the pairs $P(x_i \mid y_j)$ and $P(y_j \mid x_i)$ are called conditional entropies $H(Y \mid X)$and $H(X \mid Y)$, and given by:

$$H(Y \mid X) = -\sum_{j=1}^{m}\sum_{i=1}^{n} P(x_i, y_j)\, log_2 P(y_j \mid x_i) \qquad bits/symbol$$

Return to first equation, we have: $P(x_i, y_j) = P(x_i)P(y_j \mid x_i)$, put inside log term

$$H(X,Y) = -\sum_{j=1}^{m}\sum_{i=1}^{n} P(x_i, y_j) \log_2 P(x_i) - \sum_{j=1}^{m}\sum_{i=1}^{n} P(x_i, y_j) \log_2 P(y_j \mid x_i)$$

But

$$\sum_{j=1}^{m} P(x_i, y_j) = P(x_i)$$

Put it in above equation yields:

$$H(X,Y) = -\sum_{i=1}^{n} P(x_i) \log_2 P(x_i) - \sum_{j=1}^{m}\sum_{i=1}^{n} P(x_i, y_j) \log_2 P(y_j \mid x_i)$$

$$H(X,Y) = H(X) + H(Y \mid X)$$

2.5.3 Relationship between joint, conditional

The naturalness of the definition of joint entropy and conditional entropy is exhibited by the fact that the entropy of a pair of random variables is the entropy of one plus the conditional entropy of the other. This is proved in the following theorem which is called **(*Chain rule*):**

H(Y|X) = H(X,Y) - H(X)

H(X|Y) = H(Y,X) - H(Y)

Where: H(X|Y) is the losses entropy.

H(X,Y) is system entropy .

H(Y|X) is noise entropy.

Example 2.8: Let (X, Y) have the following joint distribution: find H(X|Y), H(y|x)

	x1	*x2*	*x3*	*x4*
y1	*1/8*	*1/16*	*1/32*	*1/32*
y2	*1/16*	*1/8*	*1/32*	*1/32*
y3	*1/16*	*1/16*	*1/16*	*1/16*
y4	*1/4*	*0*	*0*	*0*

Sol:

The marginal distribution of *X* is *(*1/2, 1/4, 1/8, 1/8*)* and

the marginal distribution of *Y* is *(*1/4, 1/4, 1/4, 1/4*)*, and hence

H(X) =-$\sum$ p(x)logp(x) = 7/4 bits and

H(Y)= $\sum$ p(y)logp(y) = 2 bits.

$$H(x,y) = \sum\sum p(x,y)logp(x,y)$$

$$= -\frac{1}{8} * \log\left(\frac{1}{8}\right) - \frac{1}{16}\log\left(\frac{1}{16}\right) - \frac{1}{32}\log\left(\frac{1}{32}\right) - \frac{1}{32}\log\left(\frac{1}{32}\right) - \frac{1}{16}\log\left(\frac{1}{16}\right)$$

$$- \frac{1}{8}\log\left(\frac{1}{8}\right) - \cdots \ldots . - 0log0 = \frac{27}{8} bit/symbol$$

H(x|y)=H(x,y) - H(y)=27/8 -2=11/8 $bit/symbol$

H(y|x) = H(x,y) - H(x) = $\frac{27}{8} - \frac{7}{4} = \frac{13}{8} = 1.625$ $bit/symbol$

Example 2.9: let $p(x_0, y_0) = \frac{1}{3}, p(x_0, y_1) = \frac{1}{3}, p(x_1, y_0) = 0, p(x_1, y_1) = \frac{1}{3}$,

find: a- H(x) , H(y) b- H(x,y)

sol:

$$a-\ \mathrm{H(x)} = \frac{2}{3} * \log\left(\frac{3}{2}\right) + \frac{1}{3} * \log(3) = log\ 3 - \frac{2}{3} = 0.918\ \text{bit/symbol},$$
$$= H(y)$$

b- $\mathrm{H(x,y)} = 3 * \frac{1}{3} * \log(3) = \log(3) = 1.585 \text{bit/symbol},$

2.6 Trans-information (Mutual information) for noisy channel

Consider the set of symbols $x_1, x_2, \ldots, x_n$, the transmitter T_x my produce. The receiver R_x may receive $y_1, y_2 \ldots\ldots\ldots y_m$. Theoretically, if the noise and jamming effect on the channel, then the set X=set Y.as shown in figure.

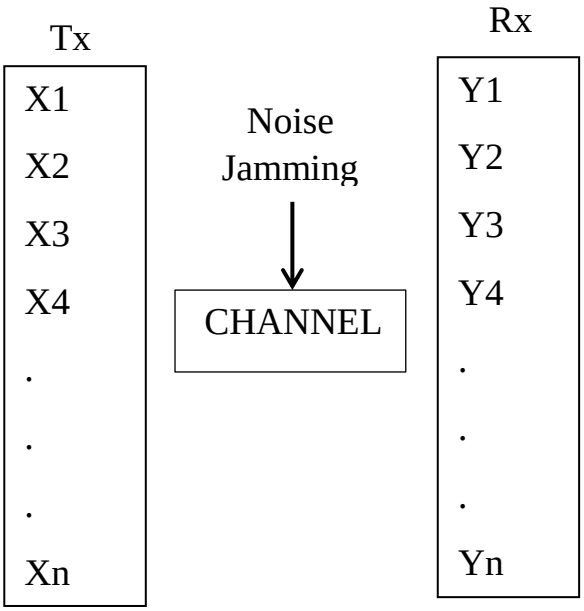


figure (2.4): The noisy channel

However and due to noise and jamming, there will be a conditional prob $P(y_j \mid x_i)$:

1- $P(x_i)$ to be what is so called the apriori prob of the symbol x_i, which is the prob of selecting x_i for transmission.

2- $P(y_j \mid x_i)$ to be what is called the aposteriori prob of the symbol x_i after the reception of y_j.

The amount of information that y_j provides about x_i is called the mutual information between x_i and y_i. This is given by:

$$I(x_i, y_j) = \log_2\left(\frac{\text{aposterori prob}}{\text{apriori prob}}\right) = \log_2\left(\frac{P(y_j \mid x_i)}{P(y_i)}\right)$$

Properties of $I(x_i, y_j)$:

1- It is symmetric, $I(x_i, y_j) = I(y_j, x_i)$.

2- $I(x_i, y_j) > 0$ if aposteriori prob> apriori prob, y_j provides +ve information about x_i.

3- $I(x_i, y_j) = 0$ if aposteriori prob= apriori prob, which is the case of statistical independence when y_j provides no information about x_i.

4- $I(x_i, y_j) < 0$ if aposteriori prob< apriori prob, y_j provides -ve information about x_i, or y_j adds ambiguity.

Also $I(x_i, y_j) = \log_2\left(\frac{P(x_i|y_j)}{P(x_i)}\right)$

2.7 Trans-information (average mutual information)

It is the statistical average of all pair $I(x_i, y_j)$, $i = 1, 2, \ldots, n,\ j = 1, 2, \ldots, m$.

This is denoted by $I(X, Y)$ and is given by:

$$I(X, Y) = \sum_{i=1}^{n}\sum_{j=1}^{m} I(x_i, y_j)P(x_i, y_j)$$

$$I(X, Y) = \sum_{i=1}^{n}\sum_{j=1}^{m} P(x_i, y_j)\log_2\left(\frac{P(y_j \mid x_i)}{P(y_j)}\right)\frac{\text{bits}}{\text{symbol}}$$

or

$$I(X,Y) = \sum_{i=1}^{n}\sum_{j=1}^{m} P(x_i, y_j) \log_2 \left(\frac{P(x_i \mid y_j)}{P(x_i)} \right) \text{ bits/symbol}$$

Expand above equation:

$$I(X,Y) = \sum_{i=1}^{n}\sum_{j=1}^{m} P(x_i, y_j) \log_2 \left(P(x_i \mid y_j) \right) - \sum_{i=1}^{n}\sum_{j=1}^{m} P(x_i, y_j) \log_2 (P(x_i))$$

And we have

$$\sum_{j=1}^{m} P(x_i, y_j) = p(x_i)$$

And by substituting:

$$I(X,Y) = \sum_{i=1}^{n}\sum_{j=1}^{m} P(x_i, y_j) \log_2 \left(P(x_i \mid y_j) \right) - \sum_{i=1}^{n} P(x_i) \log_2 (P(x_i))$$

Or $I(X,Y) = H(X) - H(X \mid Y)$

Similarly $I(X,Y) = H(Y) - H(Y \mid X)$

Thus, the mutual information of a random variable with itself is the entropy of the random variable. This is the reason that entropy is sometimes referred to as *self-information*. Collecting these results, we have the following theorem.

2.7.1 Relationship between joint, conditional and trans-information

I (X,Y) = H(X) − H(X|Y)

I (X,Y) = H(Y) − H(Y|X)

I (X, Y) = H(X) + H(Y) − H(X,Y)

I (X; Y) = I (Y; X)

I (X; X) = H(X).

Example 2.10: Show that I(X, Y) is zero for extremely noisy channel.

Sol:

For extremely noisy channel, then $y_j$gives no information about x_i the receiver can't decide anything about x_i as if we transmit a deterministic signal x_i but the receiver receives noise like signal y_j that is completely has no correlation with x_i. Then x_i and y_j are statistically independent so that $P(x_i \mid y_j) = P(x_i)$and $P(y_j \mid x_i) = P(x_i)$ for all i and j, then:

$I(x_i, y_j) = \log_2 1 = 0$ for all i & j, then $I(X, Y) = 0$ bit/symbol

Example 2.11: (Compound Probabilistic Choice System) Given a compound system of independent choices P(x,y) = {0.5, 0.1; 0.3, 0.1}, find the Mutual information

sol:

I (X; Y) = H(X) + H(Y) − H(X,Y)

Applying the law gives us the two marginal distributions Px = {0.6, 0.4} and Py = {0.8, 0.2}. We are now able to compute the entropies

$H(X, Y) = -\sum p(x,y) log p(x,y)$

$= -0.5 \cdot \log 0.5 - 0.1 \cdot \log 0.1 - 0.3 \cdot \log 0.3 - 0.1 \cdot \log 0.1 =$ 1.6855bit/symbol

$H(X) = -\sum p(x) \log p(x)$

$= -0.6 \log 0.6 - 0.4 \log 0.4 = 0.9710$ bit/symbol

$H(Y) = -\sum p(y) \log p(y)$

$= -0.8 \log 0.8 - 0.2 \log 0.2 = 0.7219$ bit/symbol,.

I (X; Y) = H(X) + H(Y) − H(X,Y)

=0.9710+0.7219-1.6855=0.0074 bit/symbol

H.W: find mutual information for this example using conditional entropy $I(X;Y) = H(X) - H(X|Y)$

Example 2.12: The joint probability of a system is given by:

$$P(X,Y) = \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix}\begin{bmatrix} 0.5 & 0.25 \\ 0 & 0.125 \\ 0.0625 & 0.0625 \end{bmatrix}$$

Find:

1- Marginal entropies. 2- Joint entropy

3-Conditional entropies. 4- The mutual information between x_1 and y_2.

5- The trans-information. 6- Draw the channel model.

Sol:

1- Marginal entropies

$$P(X) = \begin{bmatrix} x_1 & x_2 & x_3 \\ 0.75 & 0.125 & 0.125 \end{bmatrix}, \quad P(Y) = \begin{bmatrix} y_1 & y_2 \\ 0.5625 & 0.4375 \end{bmatrix}$$

$$H(X) = -[0.75\ln(0.75) + 2 \times 0.125\ln(0.125)]/\ln 2 = 1.06127 \text{ bits/symbol}$$

$$H(Y) = -[0.5625\ln(0.5625) + 0.4375\ln(0.4375)]/\ln 2$$
$$= 0.9887 \text{ bits/symbol}$$

2- Joint entropy

$$H(X,Y) = -\sum_{j=1}^{m}\sum_{i=1}^{n} P(x_i, y_j)\log_2 P(x_i, y_j)$$

$$H(X,Y)$$
$$= -\frac{[0.5\ln(0.5) + 0.25\ln(0.25) + 0.125\ln(0.125) + 2 \times 0.0625\ln(0.0625)]}{\ln 2}$$
$$= 1.875 \quad \text{bits/symbol}$$

3- Conditional entropies.

$$H(Y \mid X) = H(X,Y) - H(X) = 1.875 - 1.06127 = 0.813 \ \frac{\text{bits}}{\text{symbol}}$$

$$H(X \mid Y) = H(X,Y) - H(Y) = 1.875 - 0.9887 = 0.886 \text{ bits/symbol}$$

3-The mutual information between x_1 and y_2

$$I(x_1, y_2) = \log_2\left(\frac{P(x_1|y_2)}{P(x_1)}\right), \text{ but } P(x_1 \mid y_2) = P(x_1, y_2)/P(y_2)$$

$$I(x_1, y_2) = \log_2\left(\frac{P(x_1,y_2)}{P(x_1)P(y_2)}\right) = \log_2 \frac{0.25}{0.75 \times 0.4375} = -0.3923 \text{ bits}$$

5-That means y_2 gives ambiguity about x_1

$$I(X,Y) = H(X) - H(X \mid Y) = 1.06127 - 0.8863 = 0.17497 \text{ bits/symbol.}$$

6-To draw the channel model, must find P(Y|X) matrix from P(X, Y) matrix by dividing its rows by the corresponding $P(x_i)$:

$$P(X \mid Y) = \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix}\begin{bmatrix} 0.5/0.75 & 0.25/0.75 \\ 0/0.125 & 0.125/0.125 \\ 0.0625/0.125 & 0.0625/0.125 \end{bmatrix} = \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix}\begin{bmatrix} 2/3 & 1/3 \\ 0 & 1 \\ 0.5 & 0.5 \end{bmatrix}$$

2.8 Venn Diagrams of Entropies

The **Venn diagram**s is a helpful mean to understand the relations between mutual information and conditional entropies as shown below:

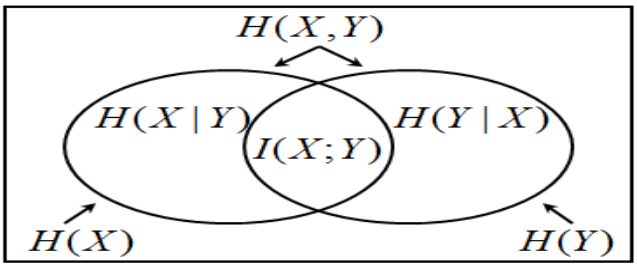


figure (2.5): relations between mutual information and conditional entropies

Problems

1-A fair die is thrown, find the amount of information gained if you are told that 4 will appear.

2-A biased coin has P(Head)=0.3. Find the amount of information gained if you are told that a tail will appear.

3-A box contains 5 green pencils and 7 yellow pencils. Two pencils are chosen at random from the box without replacement. What is the probability they are different colors?

4-We toss a fair coin three successive times. We wish to find the conditional probability P(A | B) when A and B are the events A = {more heads than tails come up}, B = {1st toss is a head}.

5-The four symbols A, B, C, D occur with probabilities(1/2 , 1/4 , 1/8, 1/8) respectively. Find the information in 3 symbols message X = BDA. Assuming that the symbols are independents.

6-A source produces dots"." and dashes"-", if the time duration of a dash is twice as long as dot and has ½ probability of a dot , if a dot intervals is 10ms. What is the average rate of information transmission?

7-A source produces dots"." and dashes"-", with P(dot) = 0.65. If the time duration of a dot = 2ms and for dash 8ms. Find average source entropy rate.

8- A two independent variables; dot (.) with probability of 0.35 and dash(-). If the duration of dot symbol is 2 msec and dash symbol duration is 3 times dot duration. Determine:

a-The entropy of the source.

b- The average source entropy rate.

9-A binary data source has P(0T) = 3/8, due to interference of channel, then P(0R/1T) = 1/16 and P(1R/0T) = 3/4. Calculate the noise and losses entropies and compare with source entropy.

10-A) Having the text (A A A B A C B C C B B A A A D B B B C D).Find

1. Find The text probability P(x) and Entropy H(x).
2. The source information of A and B(I(A), I(B))

B) If τ(A) = τ(B) = τ(C) = 0.1 μsec and τ(D) = 0.2 μsec. Calculate average Source Entropy Rate R(x).

11-A) Having the text (* * * ♯ * $ ♯ $ $ ♯ ♯ * * * & ♯ ♯ ♯ $ &) .Find:

1-The text probability P(x) and Entropy H(x).

2-The source information of A and B(I($), I(&))

B) If τ(*) = τ(♯) = τ($) = 0.1 μsec and τ(D) = 0.2 μsec. Calculate average source entropy rate R(x).

12-Eight green and ten red balls are put in a box. Two balls are chosen at random from the box without replacement.

a-Using tree diagram to determine the probability of same colors are chosen.

b-Define the entropy. What are the proprieties of mutual information$I(x_i, y_j)$.

c-What is mean by '**Nat**' and '**Hartley**', where are they used?

13- Find the entropy of the source producing the following messages:

$$Px_1 = 0.25, \quad Px_2 = 0.1, \; Px_3 = 0.15, \; and \; Px_4 = 0.5$$

14-An event has total six outcomes with probability P1=1/2, P2=1/4, P3=1/8, P4=1/16, P5=1/32, and P6=1/32. Find the entropy of the system. Also find the rate of information if these are 18 outcomes per second.

15-A transmitter has an alphabet of four letters [x1, x2, x3, x4] and the receiver has an alphabet of three letters. Calculate all entropies and transinformation if the joint probability matrix is:

$$P(X,Y) = \begin{bmatrix} 0.3 & 0.05 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0.15 & 0.05 \\ 0 & 0.05 & 0.15 \end{bmatrix}$$

16-Find and plot the entropy of binary source.

17-A) What are the entropies of a binary source if $P(0) = 0.2, 0.3$ and 0.5.

B) A system having the following joint probability matrix:

$$P(X,Y) = \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} \begin{bmatrix} 0.0625 & 0.0625 \\ 0 & 0.125 \\ 05 & 0.25 \end{bmatrix}$$

Determine:

1. Marginal entropy.
2. Source entropy.
3. Joint entropy.
4. Conditional entropy.

18-The joint probability of a system is given by: $P(X,Y) = \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} \begin{bmatrix} 0.5 & 0.25 \\ 0 & 0.125 \\ 0.0625 & 0.0625 \end{bmatrix}$

Find:

1- Marginal entropies. 2- Joint entropy

3- Conditional entropies. 4- The mutual information between x_1 and y_2.

5-The transinformation. 6- Draw the channel model.

19-The transition matrix of channel is given by

$$P(Y/X) = \begin{bmatrix} 0.1 & 0.3 & 0.6 \\ 0.2 & 0.25 & 0.55 \end{bmatrix}$$

If P(xi) = 0.4 ,find:

a) Noise entropy.

b) Losses entropy.

c) Rate of information transmission if each symbol last 0.1ms

d) Source entropy rate.

20-The joint probability of channel is defined as:

$P(x_1, y_1) = 1,\ P(x_1, y_2) = 0,\ P(x_2, y_1) = 0.5,\ P(x_2, y_2) = 0.5,$

$P(x_3, y_1) = 0.3,\quad P(x_3, y_2) = 0.7$. Find:

2- Marginal entropies.

3- Joint entropy

4- Conditional entropies.

d- The mutual information between x_1 and y_2.

e- The trans information.

f- Draw the channel model.

21-The channel model shown $P(Y/X) = \begin{bmatrix} 0.8 & 0.1 & 00.1 \\ 0.1 & 0.2 & 0.7 \\ 0.3 & 0.6 & 0.1 \end{bmatrix}$

The source information $I(X_1) = 2$ and $I(X_2) = 4$. Calculate the noise and loss entropy (H(Y/X)) and (H(X/Y)).

22-We have the joint probability of a system $P(X,Y) = \begin{bmatrix} 0.5 & 0.25 \\ 0 & 0.125 \\ 0.0625 & 0.0625 \end{bmatrix}$

Find:

a) Marginal entropies.

b) Joint entropies.

c) Conditional entropies.

d) The mutual information between x_1 and y_2.

e) The transinformation.

f) Draw the channel model.